

TACHS Exam Simulator

Full-Length Practice Exam 05

Complete printable test with 200 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Reading	50	40 minutes
Written Expression	50	40 minutes
Mathematics	50	45 minutes
Ability	50	35 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 160 minutes. Calculators should be used only when permitted by the rules followed for your TACHS preparation.

Section 1: Reading

50 questions | Recommended time: 40 minutes

Literary Passage: Before the Wind Changed

Eli had spent three weeks building a lightweight glider for the school's distance challenge. During practice, it had crossed the gym farther than any other model. At the outdoor event, however, each launch curved sharply toward the fence. Eli blamed the glider and reached for a strip of clay to weigh down its nose.

His teammate, Suri, tied short pieces of ribbon to three stakes beside the launch line. The ribbons did not point in the same direction. Those nearest the ground fluttered toward the track, while the highest one pulled toward the fence.

"The wind is turning as it rises," Suri said. She suggested shifting the launch line ten steps and waiting between gusts. Eli hesitated. Other teams were already recording distances, and changing position felt like admitting that his careful design had failed. Still, he removed the clay before pressing it onto the glider.

At the new spot, the ribbons streamed almost straight down the field. Suri counted several calm seconds after a gust. Eli released the glider, which dipped once and then sailed beyond their best indoor mark.

As they measured the flight, Eli slipped the unused clay back into the supply box. He had arrived prepared to adjust the machine. Instead, he had learned to examine the air around it.

1. What do the ribbons reveal?

- A) The launch stakes are placed too close together.
- B) The wind moves in different directions at different heights.
- C) The glider is heavier on one side than the other.
- D) The strongest gusts come from the track.

2. Why does Eli hesitate to move the launch line?

- A) He doubts that Suri understands how ribbons work.
- B) He worries that moving will damage the glider.
- C) He thinks the judges may disqualify their team.
- D) He feels that changing the conditions might make his design seem unsuccessful.

3. Which statement best expresses a central idea of the passage?

- A) Competitions reward teams that act before their opponents.
- B) Good problem solving includes examining conditions beyond the object that seems to be failing.
- C) Indoor practice is more dependable than outdoor testing.
- D) Simple materials usually improve a machine more than careful design does.

4. As used in the final paragraph, "slipped" most nearly means

- A) lost his balance near.
- B) moved accidentally from.
- C) placed quietly into.
- D) escaped unnoticed through.

Informational Passage: Fences That Help Dunes Grow

Along some coastlines, rows of low wooden slats stand several feet inland from the water. These sand fences are not intended to stop waves. Instead, they slow wind near the ground. When wind carrying grains of sand passes through the gaps, it loses some speed and drops part of its load. Over time, the deposited sand forms a ridge. Placement is important. A fence set too close to regular high water may be washed away before a dune develops. A fence with no gaps may force wind over its top at high speed, causing erosion on the other side. For this reason, sand fences are usually porous: air passes through, but more slowly.

Once a small mound forms, coastal grasses can strengthen it. Their stems trap additional windblown sand, while their spreading roots help hold the growing dune together. Managers often protect these plants by directing visitors to raised walkways.

Sand fences cannot protect every shore. They work best where wind routinely carries dry sand inland and where enough space exists for dunes to form. They also require monitoring after storms. Used under suitable conditions, however, the fences begin a process that wind and vegetation continue. The finished dune is not built board by board; it develops because the boards change where moving sand comes to rest.

5. Why are sand fences usually built with gaps?

- A) Gaps allow plants to grow through the slats immediately.
- B) Gaps keep waves from reaching the back of the fence.
- C) Gaps make the fence easy to move after each storm.
- D) Gaps slow the air without forcing all of it rapidly over the fence.

6. How is the passage mainly organized?

- A) It presents two shoreline problems and argues that only one can be solved.
- B) It explains how a tool begins a natural process, identifies design conditions, and describes limitations.
- C) It traces coastal construction methods from ancient to modern times.
- D) It gives instructions for building a fence and calculating its cost.

7. Why does the author mention raised walkways?

- A) To suggest that tourists should be able to view storms safely
- B) To compare wooden slats with stronger building materials
- C) To explain why dunes must form close to regular high water
- D) To show one way managers protect plants that help stabilize dunes

8. Which coastline would most likely benefit from sand fences?

- A) A broad beach where steady winds carry dry sand inland
- B) A sheltered harbor where wind rarely reaches the ground
- C) A narrow rocky cliff where little loose material is present
- D) A concrete seawall that is covered by every high tide

Humanities or Reflective Passage: The Pause in an Interview

Beginning interviewers often fear silence. As soon as a speaker finishes a sentence, the interviewer may rush to ask the next prepared question. The conversation continues smoothly, but useful thoughts may remain unspoken. A pause can serve as part of a question. A person describing an old neighborhood, for example, may stop because a name or detail has not yet returned to mind. If the interviewer waits, the speaker may add a memory that no checklist could have predicted. Silence can also give the speaker time to decide how to express an experience accurately.

Waiting does not mean becoming passive. A thoughtful interviewer watches for signs that distinguish reflection from discomfort. A relaxed expression or a glance upward may suggest that the speaker is searching for words. A tense posture or repeated glance toward the door may signal that the subject should change. Respectful listening requires attention to both possibilities.

Prepared questions still matter. They establish the interview's purpose and ensure that essential topics are addressed. Yet the questions are a framework, not a script that must control every second. The interviewer's task is partly to ask and partly to make room. In that room, a speaker may move from giving information to revealing what the information means.

9. Which statement best summarizes the passage?

- A) Interviewers should replace prepared questions with informal conversation.
- B) Speakers communicate most clearly when interviews follow a strict schedule.
- C) Effective interviewing balances preparation with attentive, respectful pauses.
- D) Silence is useful only when a speaker cannot remember a name.

10. According to the passage, what may a relaxed expression or upward glance indicate?

- A) The prepared questions are too difficult.
- B) The speaker wants the interview to end.
- C) The interviewer has asked an irrelevant question.
- D) The speaker is searching for words.

11. As used in the final paragraph, "framework" most nearly means

- A) a permanent rule that cannot be changed.
- B) a basic structure that guides the conversation.
- C) a record of everything a speaker says.
- D) a barrier separating two people.

12. What does the author imply about an interviewer who asks questions too quickly?

- A) The interviewer will always make the speaker uncomfortable.
- B) The interviewer probably has not prepared enough questions.
- C) The interviewer may collect facts while missing deeper reflections.
- D) The interviewer is likely to forget the interview's main purpose.

13. The author's tone is best described as

- A) practical and thoughtful.
- B) humorous and playful.
- C) doubtful and impatient.
- D) formal and disapproving.

Literary Passage: The Two-Minute Scene

For the drama club's festival, Tessa had written a two-minute scene about travelers waiting for a train. Because she could picture every movement, she demonstrated exactly where each actor should stand, when a suitcase should close, and how loudly every line should be spoken. Rehearsal was orderly, but the scene felt longer than two minutes.

During a break, Andre quietly tried balancing his hat on the suitcase. When the lid sprang open, the hat flew onto Lila's head. She answered his surprised expression with a serious announcement that the "train" had changed platforms. The other actors laughed, then repeated the exchange without speaking. It took only a few seconds and made the travelers' confusion immediately clear.

Tessa's first impulse was to restore her original directions. Instead, she asked the group to show her one movement each character might contribute. They kept the hat accident, cut three explanatory lines, and created a quick sequence in which the actors crossed paths while following conflicting signs.

At the festival, the audience understood the problem before anyone spoke. The final scene lasted one minute and fifty-two seconds. Tessa still recognized her story, but it now carried ideas she could not have supplied alone.

14. What happens after Andre balances his hat on the suitcase?

- A) Lila adds three explanatory lines.
- B) Tessa ends the rehearsal early.
- C) The actors move the scene to another room.
- D) The suitcase lid sends the hat onto Lila's head.

15. Why does the original scene probably feel too long?

- A) Too much is explained verbally instead of shown efficiently.
- B) The suitcase is difficult to open on cue.
- C) The actors have not memorized their lines.
- D) Tessa has written a story with several different settings.

16. Which statement best expresses a central idea of the passage?

- A) A writer should avoid changing a script after rehearsal begins.
- B) Strict directions are necessary when time is limited.
- C) Silent scenes are always more effective than spoken scenes.
- D) Shared creativity can strengthen a leader's original idea.

17. As used in the final sentence, "carried" most nearly means

- A) transported by hand.
- B) continued for a distance.
- C) included and communicated.
- D) supported the weight of.

Scientific or Technical Passage: When Lake Ice Sings

On a cold, still day, a frozen lake may produce a sharp ping, a low boom, or a sound that seems to sweep across the ice. These noises often result from changes in temperature rather than from something striking the surface. Ice expands and contracts as its temperature changes. When the surface cools rapidly, the upper layer contracts sooner than the ice beneath it. Stress builds until a crack forms or lengthens. That sudden movement sends vibrations through both the ice and the water below. Because different vibration frequencies travel at different speeds, a nearby listener may hear a descending, science-fiction-like tone.

The sound alone does not reveal whether the ice can safely support a person. Ice strength depends on thickness, recent weather, currents, springs, snow cover, and other conditions. A loud boom may come from thick ice adjusting to cold, while dangerously thin ice may remain quiet. For that reason, people should rely on local safety guidance and direct measurements, not on sound.

Researchers record these noises to study how cracks begin and travel. The recordings also demonstrate that a frozen surface is not motionless. Even when a lake appears perfectly still, temperature is continually changing the forces within its ice.

18. What directly produces vibrations when lake ice cools rapidly?

- A) Snow pressing evenly on the surface
- B) A crack forming or extending as stress is released
- C) Air moving through openings beneath the ice
- D) A current repeatedly striking the shore

19. How is the passage mainly organized?

- A) It describes a phenomenon, explains its cause, corrects a possible misunderstanding, and notes a research use.
- B) It compares several lakes and ranks them by safety.
- C) It gives chronological instructions for crossing a frozen lake.
- D) It presents two theories and concludes that neither has evidence.

20. Why does the author contrast loud, thick ice with quiet, thin ice?

- A) To show that thicker ice always produces lower tones
- B) To prove that snow cover prevents ice from cracking
- C) To explain why researchers prefer studying thin ice
- D) To emphasize that sound is not a dependable measure of safety

21. Which statement best summarizes the passage?

- A) Temperature-driven cracks can make ice sound unusual, but the sounds do not establish whether the ice is safe.
- B) Scientists can determine ice thickness by comparing its tones.
- C) Lake sounds mainly come from objects falling onto weak ice.
- D) Frozen lakes become completely motionless once their surfaces appear solid.

Informational or Historical-Style Passage: Tokens Inside Clay Envelopes

Long before paper receipts, some communities in ancient Mesopotamia used small clay tokens to represent goods. A cone might stand for a measure of grain, while a different shape could represent an animal or another product. When people recorded a transaction, they placed the appropriate tokens inside a hollow clay ball and sealed it. The sealed container is often called a bulla.

The system protected the tokens, but it created a difficulty: once the clay hardened, no one could see what was inside without breaking the container. To solve this problem, people pressed the tokens into the wet outer surface before sealing them inside. The impressions created a visible record of the contents.

Eventually, record keepers discovered that marks on a flat clay tablet could preserve the needed information without enclosing the objects themselves. They began drawing or pressing symbols directly into tablets. This change did not produce a complete writing system at once, but it illustrates an important shift: a physical object that represented a quantity was replaced by a mark that represented that object.

Scholars study surviving tokens, bullae, and tablets to understand this development. The evidence is incomplete, so interpretations may change when new objects are found. Even so, these materials reveal how practical accounting methods helped people move toward recording information with symbols.

22. Why were tokens pressed into the outside of a bulla?

- A) To make the container stronger after it dried
- B) To decorate the record with pictures of local products
- C) To show the contents without breaking the sealed container
- D) To identify the person who shaped the clay

23. As used in the final paragraph, "surviving" most nearly means

- A) defeating a serious danger.
- B) continuing to exist from an earlier time.
- C) functioning exactly as originally intended.
- D) being copied into a newer form.

24. What can most reasonably be inferred about the change from tokens to marks?

- A) Record keepers no longer needed to represent quantities.
- B) Information became less dependent on storing the represented objects themselves.
- C) Flat tablets were useful only for recording animals.
- D) The first marks immediately allowed people to record any spoken sentence.

25. The author mentions that interpretations may change mainly to

- A) prove that tokens had no consistent meanings.
- B) argue that clay records have little value to scholars.
- C) suggest that most surviving objects are modern copies.
- D) acknowledge that conclusions are based on incomplete evidence.

Literary Passage: The Fourth Runner

Nora was the alternate on her school's relay team. She attended every practice, but the four regular runners received the batons, lane assignments, and cheers. Nora told herself that an alternate's main job was to stay out of the way.

One afternoon, the coach asked her to observe the handoffs and record where each runner began accelerating. Nora noticed that Malik often looked back too early, slowing before the baton reached him. She also saw that June started near a faded mark that was difficult to find from the curve. Nora placed a bright strip of removable tape beside the mark and suggested that Malik listen for a single call instead of turning around.

At the next meet, June twisted her ankle while warming up. Nora had ten minutes to join the lineup. Her hands shook as she took June's position, but the taped mark was exactly where she expected it. Malik kept his eyes forward when Nora called. Their exchange was not the fastest of the day, yet it was clean.

The team finished second. While the others discussed their times, the coach handed Nora the observation sheet. It was covered with notes from every practice. Nora realized she had not begun contributing when she entered the race. She had been part of it while standing beside the track.

26. What does Nora place beside June's starting mark?

- A) A bright strip of removable tape
- B) A copy of the observation sheet
- C) A spare baton
- D) A lane-assignment card

27. Why is Nora able to enter the relay with little preparation time?

- A) She has competed in June's position at several earlier meets.
- B) The coach changes the order of all four runners.
- C) She has carefully studied the team's handoffs during practice.
- D) The other runners slow their pace for her.

28. Which statement best expresses a central idea of the passage?

- A) Athletic success depends mainly on finishing first.
- B) Alternates should always receive the same assignments as regular runners.
- C) Recording race times is more useful than practicing handoffs.
- D) Careful preparation can make a contribution valuable before it becomes visible.

29. As used in the final paragraph, "entered" most nearly means

- A) wrote information into a record.
- B) crossed through a doorway.
- C) submitted a name for consideration.
- D) began participating directly in.

Literary Passage: The Useful Wrong Turns

Priya and Mateo were designing a puzzle for the coding club's open house. Players would guide a small robot through a grid by entering commands. Priya wanted to erase every failed command sequence from their notes. "Visitors only need the route that works," she said.

Mateo kept the crossed-out attempts. He marked each place where the robot turned one square too early, stopped at the border, or repeated a move. When they invited another student to test the puzzle, she reached the goal on her first try and shrugged. "It's clear, but it isn't much of a puzzle."

Mateo spread the failed sequences across the table. Several errors had sent the robot to the same three squares. He suggested placing optional tokens on those squares. Players could take a shorter, difficult route to collect all three tokens or follow the longer route directly to the goal.

Priya rebuilt the program so that both choices worked. At the open house, some visitors competed for the quickest finish, while others tried to collect every token. The final puzzle depended on paths its designers had once labeled useless.

As they packed up, Priya did not save every mistaken command. She did, however, keep the marked sheet. It reminded her that a failed route could contain information even when it did not contain the answer.

30. What criticism does the student tester make?

- A) The puzzle is clear but not very challenging.
- B) The robot moves too slowly across the grid.
- C) The commands are too difficult to enter.
- D) The optional tokens are placed too far apart.

31. Why does Priya keep the marked sheet at the end?

- A) She now understands that mistakes can reveal useful possibilities.
- B) She plans to require visitors to copy every failed route.
- C) She wants to prove that Mateo entered the wrong commands.
- D) She needs the sheet to rebuild the robot's motor.

32. The author includes the visitors' different goals mainly to show that

- A) the robot should have moved at several speeds.
- B) the revised puzzle supports more than one satisfying way to play.
- C) most visitors misunderstand the designers' instructions.
- D) collecting tokens is easier than reaching the goal.

33. How does the tester's comment affect the plot?

- A) It ends the disagreement by proving Priya's plan was correct.
- B) It introduces a new character who takes control of the project.
- C) It creates a problem that leads the designers to reconsider their discarded attempts.
- D) It causes the club to cancel the open-house demonstration.

Informational Passage: Catching Water from Fog

Fog consists of tiny water droplets suspended in air. In some windy, foggy regions where rainfall is limited, large mesh panels can collect a portion of those droplets. These structures are called fog collectors or fog nets. As moist air passes through a net, droplets strike its fibers and join with other droplets. Once enough water gathers, gravity pulls it downward. A channel beneath the mesh directs the water into a storage container. The process requires no pump to move water through the net, although pipes or pumps may later carry the collected water elsewhere.

Mesh design affects performance. Fibers must provide surfaces for droplets to hit, but the net cannot block so much air that wind flows around it. The net's height, direction, and location also matter. A panel facing frequent fog-bearing winds may collect far more than an identical panel placed behind a hill.

Fog nets do not create water, and their output changes with weather. They are therefore unlikely to replace every other water source. In suitable locations, however, they can provide water for tree seedlings, gardens, or limited household uses. Their value comes from matching a simple device to a particular local pattern of moving air.

34. What moves collected droplets from the mesh into the channel?

- A) Gravity
- B) A pump inside the panel
- C) Heat from the storage container
- D) Pressure from falling rain

35. As used in the third paragraph, "performance" most nearly means

- A) how effectively the net collects water.
- B) the way actors complete a scene.
- C) a public presentation.
- D) the strength of the wind behind a hill.

36. Which statement best summarizes the passage?

- A) Fog nets turn dry air into unlimited drinking water.
- B) Storage containers are more important than the mesh used above them.
- C) Fog nets collect airborne droplets through carefully positioned mesh, but their usefulness depends on local conditions.
- D) Fog nets work best when their fibers prevent all air from passing.

37. Which proposed site would most likely collect the greatest amount of fog water?

- A) An exposed ridge facing regular fog-bearing winds
- B) A flat indoor room with high humidity
- C) A dry slope facing away from the usual wind
- D) A sheltered valley where air usually remains still

38. Why does the author state that fog nets do not create water?

- A) To argue that the nets have no practical use
- B) To clarify a limitation and prevent an exaggerated understanding of the technology
- C) To show that rainfall always supplies more water than fog
- D) To explain why every fog net requires an electric pump

Literary Passage: The Unclaimed Sound

For the school play, Lena managed a table crowded with objects used to create sound effects. A metal tray became thunder, dried beans in a box became rain, and two coconut shells became hoofbeats. During rehearsal, however, the director stopped a scene set in an old train station. The recorded station noises sounded distant and flat.

Lena searched the table, but none of its labeled objects seemed useful. Beside the stage door sat an unclaimed bicycle wheel. She turned it slowly and heard a faint clicking from a card caught between the spokes. When she held a sheet of stiff paper against the spinning spokes, the clicks became a steady mechanical rhythm. Adding a short hiss from a spray bottle made the sound resemble a train gathering speed.

The director asked Lena to repeat the effect during the scene. She practiced changing the wheel's speed to match the actors' movements. On performance night, the audience never saw the bicycle wheel, yet several people later praised the convincing departure.

As Lena returned the props, she made a new label for the wheel. Then she paused and wrote it in pencil. The wheel had taught her that an object's first use did not have to be its last.

39. What does Lena add to the wheel sound to suggest a moving train?

- A) Two coconut shells tapped together
- B) Dried beans shaken in a box
- C) A strike on a metal tray
- D) A short hiss from a spray bottle

40. Why does Lena write the wheel's new label in pencil?

- A) She expects the director to erase every prop label.
- B) She is unsure how to spell the name of the object.
- C) She wants to leave room for another possible use.
- D) She plans to return the wheel before the next scene.

41. The description of the audience praising a sound source they never saw mainly emphasizes

- A) how easily an audience can identify ordinary objects.
- B) the effectiveness of Lena's inventive backstage work.
- C) why recorded sounds are always unsuitable for plays.
- D) the director's failure to explain the scene.

42. As used in the first paragraph, "flat" most nearly means

- A) level in shape.
- B) without a raised surface.
- C) unable to roll.
- D) lacking realism or energy.

Informational Passage: Why Some Seeds Wait

A seed may appear inactive, but conditions inside and around it determine when growth begins. Many seeds remain dormant until they receive particular signals. This delay can prevent a young plant from emerging during a season when it is unlikely to survive.

Some seeds have coats that water cannot easily penetrate. Scraping by soil, passage through an animal's digestive system, or repeated freezing and thawing may weaken the coat. Only then can water enter and start chemical processes needed for growth. Other seeds respond to temperature. A period of winter cold can prepare them to sprout when warmer spring conditions arrive.

Fire provides a signal for certain plants in fire-prone habitats. Heat may crack a hard seed coat, while chemicals in smoke can trigger changes in other species. These adaptations do not mean every seed germinates after every fire. The intensity and duration of heat still matter, and seeds buried at different depths experience different conditions.

Dormancy spreads risk over time. If every seed from a plant sprouted after the first warm day, one late frost might destroy the entire group. When seeds respond to different combinations of signals, at least some may begin growing under favorable conditions.

43. According to the passage, what can weaken a water-resistant seed coat?

- A) Repeated freezing and thawing
- B) Moonlight reflected from snow
- C) Shade from a mature tree
- D) Pollen carried by insects

44. Which statement best summarizes the passage?

- A) All seeds require fire before water can enter their coats.
- B) Seed dormancy delays growth until particular signals improve the chance of survival.
- C) Seeds buried deeply always sprout earlier than surface seeds.
- D) Winter cold prevents seeds from undergoing chemical changes.

45. As used in the final paragraph, “favorable” most nearly means

- A) unusual.
- B) suitable for success.
- C) permanent.
- D) easy to observe.

46. Why does the author mention a late frost in the final paragraph?

- A) To prove that warm days never occur before spring ends
- B) To show that all dormant seeds survive severe weather
- C) To illustrate the risk of an entire group sprouting at once
- D) To explain why buried seeds receive less heat

Biographical-Style Passage: Maps Made by Listening

When city planner Amara Reed began studying a busy neighborhood intersection, she first examined traffic counts and street maps. The records showed where cars turned and how long signals remained green. They did not show why residents avoided the small plaza beside the intersection.

Reed organized short walks with people who used the area differently. A parent pushing a stroller pointed out a curb cut that collected rainwater. A shopkeeper explained that delivery trucks blocked the plaza each morning. An older resident said the benches faced afternoon glare, while a student noticed that cyclists had no clear place to enter the nearby path.

Reed marked each observation on a transparent sheet placed over the street map. The layers did not replace the traffic data; they revealed experiences the numbers had missed. Her team proposed moving the delivery zone, adjusting two benches, improving drainage, and painting a clear bicycle connection.

Months after the changes, more people paused in the plaza. Reed continued using traffic counts, but she also made “listening walks” a regular step in later projects. To her, a map was not merely a picture of streets. It was a question about how people moved through them.

47. Which problem does the parent identify?

- A) A curb cut collects rainwater.
- B) The traffic signal changes too quickly.
- C) The bicycle path is too narrow.
- D) The shops close before afternoon.

48. What can be inferred about Reed’s view of traffic data?

- A) It should be discarded whenever residents disagree with it.
- B) It matters only when a new street is being constructed.
- C) It is useful but does not capture every relevant experience.
- D) It provides more detail than direct observation can provide.

49. What is the central idea of the passage?

- A) Public plazas should never be located near intersections.
- B) Residents usually understand traffic signals better than planners do.
- C) Combining numerical records with users’ experiences can lead to better design.
- D) Transparent maps are more accurate than printed street maps.

50. How is the passage mainly organized?

- A) A problem is identified, additional evidence is gathered, and solutions are applied.
- B) Several planners debate two competing designs.
- C) A neighborhood’s history is described in chronological periods.
- D) A theory is presented and then disproved by an experiment.

Section 2: Written Expression

50 questions | Recommended time: 40 minutes

Revision Passage: Creating a Student Audio Guide

[1] Our class is creating an audio guide for visitors to the school's local-history display. [2] Each recording should explain one object in about a minute. [3] The basketball team practices in the gym on Tuesdays. [4] Short recordings will let visitors choose several objects without spending too long at one station. [5] To make the guide useful, students should identify the object, explain why it matters, and describe one detail that visitors might otherwise miss. [6] Students will record their scripts in the media room, the room has sound-absorbing panels. [7] We should also test the guide with people who have not studied the display. [8] Their questions can reveal where our explanations need more context.

51. Which sentence should be deleted because it does not support the passage's central idea?

- A) Sentence 2
- B) Sentence 3
- C) Sentence 4
- D) Sentence 7

52. Which is the best revision of sentence 6?

- A) Students will record their scripts in the media room and sound-absorbing panels.
- B) Students will record their scripts in the media room, having sound-absorbing panels.
- C) Students will record their scripts in the media room, which has sound-absorbing panels.
- D) Students will record their scripts in the media room because of the panels, they absorb sound.

53. Which revision of sentence 2 is most precise?

- A) Each recording should say some things about one object for a little while.
- B) Each recording should give visitors a focused explanation of one object in about a minute.
- C) There should be a recording that is not very long and is about an object.
- D) Visitors will listen to objects explained by recordings of approximately some time.

54. Which sentence would best follow sentence 5 and strengthen its development?

- A) Audio files can be stored in several different formats.
- B) Some students prefer history to science.
- C) For example, a speaker describing an old lunch token might explain what it cost and point out the faded school emblem.
- D) The display cases were moved from another hallway last year.

Revision Passage: Binding My First Notebook

[1] At a weekend workshop, I learned to bind a small notebook by hand. [2] After folding the paper into sections, the instructor showed us how to mark evenly spaced holes along each fold. [3] I pierced the holes with an awl and then stitched the sections together with waxed thread. [4] Pulling as hard as possible, the thread tore one of my pages. [5] On my second attempt, I tightened each stitch only until the pages rested firmly together. [6] The cover then lay flat, and the pages turned without catching. [7] My first attempt was unsuccessful and did not succeed. [8] The finished notebook showed me that careful tension matters more than force.

55. Which is the best revision of sentence 4?

- A) Pulling the thread as hard as possible, I tore one of my pages.
- B) The thread, pulling as hard as possible, tore one of my pages.
- C) Pulling as hard as possible, one of my pages was torn by the thread.
- D) One of my pages, as hard as possible, was torn by pulling the thread.

56. Which is the most concise revision of sentence 7?

- A) My first attempt was an attempt that was unsuccessful.
- B) My first attempt failed.
- C) My first attempt was not successful in succeeding.
- D) My first attempt did not have a successful result in the end.

57. Where would the following sentence best be placed?

The wax coating helped the thread slide through the holes without fraying.

- A) After sentence 1
- B) After sentence 3
- C) After sentence 6
- D) After sentence 8

58. Which sentence best combines sentences 5 and 6?

- A) I tightened each stitch only until the pages rested firmly together, the cover then laying flat, the pages turned without catching.
- B) On my second attempt, tightening each stitch, the pages rested firmly together, the cover lay flat and turning without catching.
- C) On my second attempt, I tightened each stitch only until the pages rested firmly together; as a result, the cover lay flat and the pages turned without catching.
- D) The pages rested firmly together on my second attempt and because tightening each stitch, with the cover flat.

59. Which sentence is written correctly?

- A) By the time the bus arrived, we wait for nearly thirty minutes.
- B) By the time the bus arrives, we had waited for nearly thirty minutes yesterday.
- C) By the time the bus had arrived, we are waiting for nearly thirty minutes.
- D) By the time the bus arrived, we had waited for nearly thirty minutes.

60. Which sentence uses the comparative form correctly?

- A) Of the two routes, the path beside the canal is shortest.
- B) Of the two routes, the path beside the canal is shorter.
- C) The path beside the canal is more shorter than the hill route.
- D) The path beside the canal is the most shortest of the two.

61. Which sentence is most appropriate for a formal email requesting information from a museum?

- A) Hey, tell me when your fossil room is open.
- B) I need the fossil-room hours, so send them soon.
- C) What's up with the schedule for all that fossil stuff?
- D) Would you please let me know the hours for the fossil gallery during spring break?

62. Which line contains an error in standard written US English?

Line 1: The collection of postcards was donated last month.

Line 2: Each card shows a street or landmark from the region.

Line 3: Several of the oldest cards has handwritten messages.

Line 4: Volunteers are placing them in protective sleeves.

- A) Line 1
- B) Line 2
- C) Line 4
- D) Line 3

63. Which line contains an error in standard written US English?

Line 1: Lena set the compass beside the map.

Line 2: She checked the trail markers carefully.

Line 3: Her brother had already packed the water bottles.

Line 4: Together, they laid the supplies careful into the bag.

- A) Line 4
- B) Line 2
- C) Line 3
- D) Line 1

Revision Passage: Comparing Two Water Filters

[1] Our science group compared two homemade water filters, one filled with fine sand and one filled with small gravel. [2] We poured equal amounts of cloudy water through each filter and collected the liquid in identical clear cups. [3] The sand filter produced clearer water, it required nearly twice as long to drain. [4] By contrast, water passed quickly through the gravel, but some visible particles remained. [5] These results suggest that smaller spaces between materials can trap more particles while slowing the flow. [6] The cups were placed on a table. [7] Because neither filter made the water safe to drink, our demonstration concerned filtration speed and clarity only. [8] Future trials could test layered filters while keeping the water amount and collection time consistent.

64. Which is the best revision of sentence 3?

- A) The sand filter produced clearer water, but it required nearly twice as long to drain.
- B) The sand filter produced clearer water, requiring nearly twice as long to drain, however.
- C) Producing clearer water, nearly twice as long was required by the sand filter to drain.
- D) The sand filter produced clearer water it required nearly twice as long to drain.

65. Which sentence should be deleted because it adds no useful information to the explanation?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 6
- D) Sentence 8

66. Which revision of sentence 7 is clearest and most appropriate for a science report?

- A) The water was not for drinking and speed and clarity were what it was about.
- B) You definitely should not drink this water because the filters were just homemade things.
- C) Neither filter purified the water for drinking; the test measured only filtration speed and visible clarity.
- D) Neither filter did everything, which is important to know about the whole experiment.

67. Which sentence would best follow sentence 5?

- A) Gravel is commonly sold in heavy bags at garden stores.
- B) Many experiments can be repeated by different students.
- C) The group selected clear cups instead of colored containers.
- D) This trade-off explains why the clearer sample took longer to collect.

Revision Passage: A Poetry Path for the Courtyard

[1] The literary magazine proposes creating a temporary poetry path across the school courtyard. [2] Short student poems would be printed on weather-resistant cards and attached to removable stands. [3] Visitors could follow the stands in any order, pausing wherever a title interests them. [4] To include different voices, submissions would be read without authors' names during selection. [5] The committee would choose poems for clarity, originality, and how effectively they use imagery. [6] After choosing the poems, a map of stand locations would be made by the committee. [7] The display would remain for one week, and all stands would be removed afterward. [8] This limited schedule would allow the courtyard to return promptly to its regular use.

68. Which is the best revision of sentence 6?

- A) After choosing the poems, a map would be made, being stand locations.
- B) After choosing the poems, the committee would map the locations of the stands.
- C) The poems chosen, stand locations, and the committee would make a map.
- D) A map of locations after choosing poems would be made by them.

69. Which revision of sentence 3 is most concise without changing its meaning?

- A) Visitors could explore the stands in any order and pause at appealing titles.
- B) Visitors could go to stands in an order, and titles might interest them enough to pause.
- C) The stands could be followed by visitors, who could pause at titles of interest to them.
- D) In any order that they decide upon, visitors could follow around the various stands.

70. Which sentence would provide the most effective conclusion after sentence 8?

- A) Some poems use rhyme, while others are written in free verse.
- B) The magazine publishes two regular issues during the school year.
- C) The path would turn an ordinary crossing place into a brief, shared encounter with student writing.
- D) Courtyards may contain benches, plants, pavement, and doors.

71. Which sentence best combines sentences 7 and 8?

- A) The display would remain for one week; then all stands would be removed so the courtyard could promptly return to regular use.
- B) Remaining for one week, afterward all stands would be removed, the courtyard returned to regular use.
- C) The display would remain for one week, all stands removed afterward and regular use promptly.
- D) All stands after one week, which is temporary, would be removed by the display.

72. Which sentence is written correctly?

- A) Every one of the lanterns need a fresh battery.
- B) Every one of the lanterns needing a fresh battery.
- C) Every one of the lanterns needs a fresh battery.
- D) Every one of the lanterns need fresh batteries, they are dim.

73. Which sentence uses pronouns correctly?

- A) The guide gave Omar and I a copy of the schedule.
- B) The guide gave myself and Omar a copy of the schedule.
- C) The guide gave Omar and me a copy of the schedule.
- D) The guide gave he and Omar a copy of the schedule.

74. Which line contains an error in standard written US English?

Line 1: The volunteers sorted the donated fabric by color.

Line 2: A basket near the door held smaller pieces.

Line 3: Mira chose the patterned cloth because it matched perfectly.

Line 4: She then cut the material more neat than before.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

75. Which line contains an error in standard written US English?

Line 1: Neither the strong winds nor the rain were expected.

Line 2: The crew secured every banner before sunset.

Line 3: A supervisor checked each knot twice.

Line 4: By morning, all of the equipment remained in place.

- A) Line 2
- B) Line 1
- C) Line 3
- D) Line 4

76. Which line contains an error in standard written US English?

Line 1: We had completed the inventory before the delivery arrived.

Line 2: The driver placed the boxes beside the storage-room door.

Line 3: Later, two assistants carries the boxes inside.

Line 4: The manager compared the shipment with the order form.

- A) Line 3
- B) Line 2
- C) Line 1
- D) Line 4

Revision Passage: Mapping the Sounds of the School

[1] Our environmental science class created a sound map of the school grounds. [2] At six locations, we recorded the loudest sound every five minutes during lunch. [3] Students also noted whether each sound came from people, traffic, equipment, or animals. [4] The fountain area had a lot of noise and stuff happening there. [5] Near the library wall, conversation was quieter, but delivery trucks were briefly loud. [6] Because the recordings were taken at equal intervals, comparisons between locations were more consistent. [7] The art room contains twelve sinks. [8] The map can help the school identify places suited to quiet activities and places where louder events would cause less disruption.

77. Which is the best revision of sentence 4?

- A) The fountain area was a place where various things made some noise.
- B) At the fountain, splashing water and nearby conversations produced steady background noise.
- C) The fountain area had noise, people, water, and other stuff around it.
- D) A lot was happening loudly by the fountain area in several ways.

78. Which sentence should be deleted because it is unrelated to the sound-mapping project?

- A) Sentence 2
- B) Sentence 5
- C) Sentence 7
- D) Sentence 8

79. Which sentence best combines sentences 2 and 3?

- A) At six locations, recording the loudest sound every five minutes during lunch and its source.
- B) At six locations, we recorded the loudest sound every five minutes during lunch, students noted its source.
- C) At six locations during lunch, the loudest sound was recorded every five minutes and students noting sources.
- D) At six locations, we recorded the loudest sound every five minutes during lunch and classified its source as people, traffic, equipment, or animals.

80. Which revision of sentence 8 most effectively states the project's practical value?

- A) The map can guide decisions about where to schedule quiet study and louder group events.
- B) The map contains information about sounds that were heard in locations.
- C) Quiet and loud activities are different, and the map shows this fact.
- D) The school has several places where events and other things may happen.

Revision Passage: Building a Miniature Stage Set

[1] For the design workshop, our group built a miniature stage set from foam board. [2] We first measured the model wall and marked the positions of two doors. [3] Cutting the openings with a craft knife, the ruler was used to guide each straight edge. [4] We reinforced the corners with narrow paper strips before adding paint. [5] The painted doors looked realistic, but both opened into the same small space behind the wall. [6] We therefore moved one door and added a shallow divider so each entrance suggested a different room. [7] The revision made the model clearer to an audience viewing it from the front. [8] It was a change that improved it in a better way.

81. Which is the best revision of sentence 3?

- A) We used a ruler to guide each straight edge as we cut the openings with a craft knife.
- B) Cutting the openings with a craft knife, each straight edge was guided by the ruler.
- C) The ruler, cutting the openings with a craft knife, guided each straight edge.
- D) Each straight edge with a ruler, the openings were cut by using a craft knife.

82. Which sentence would best follow sentence 5?

- A) Foam board is available in several thicknesses and colors.
- B) The workshop tables were covered with protective paper.
- C) Some full-size stages use curtains instead of solid walls.
- D) From the audience's viewpoint, that arrangement made the two entrances seem to lead to one room.

83. Which is the most concise revision of sentence 8?

- A) It was a change that made an improvement in a better manner.
- B) The change was an improved change to the model.
- C) In a better way, the model was improved by this revision.
- D) The revision improved the model.

84. Which sentence is written correctly?

- A) The committee has chose three photographs for the exhibit.
- B) The committee has choosing three photographs for the exhibit.
- C) The committee has chosen three photographs for the exhibit.
- D) The committee have chose three photographs for the exhibit.

85. Which sentence uses a pronoun with a clear antecedent?

- A) When Mia placed the glass beside the pitcher, it was nearly full.
- B) The oak tree lost several branches during the storm, but it remained standing.
- C) Carlos handed the permit to Mr. Lee after he signed it.
- D) Jenna told Ava that her folder was missing.

86. Which choice is a complete sentence?

- A) Although the final bus arrived ten minutes late.
- B) After the final bus arrived, the volunteers locked the gate.
- C) The posters beside the entrance to the auditorium.
- D) Carrying the folded chairs into the auditorium.

87. Which line contains an error in standard written US English?

Line 1: The museum displayed several tools from the workshop.

Line 2: One wooden plane still had its maker's initials.

Line 3: Visitors examined the objects respectfully and quietly.

Line 4: A guide explain how each tool had been used.

- A) Line 4
- B) Line 2
- C) Line 3
- D) Line 1

88. Which line contains an error in standard written US English?

Line 1: Before sunrise, the crew has loaded the equipment.

Line 2: The van reached the field at six o'clock.

Line 3: Everyone carried their assigned case to the tent.

Line 4: The coordinator checked the list before the first event began.

- A) Line 2
- B) Line 1
- C) Line 3
- D) Line 4

Revision Passage: A Quiet Cafeteria Corner

[1] Our student council proposed creating a quiet corner in the cafeteria for students who want to read or finish schoolwork during lunch. [2] The cafeteria is the largest room in the school. [3] At first, we planned to place the corner beside the kitchen doors. [4] After observing the room, however, we noticed carts passed that spot throughout lunch. [5] We then selected an unused area near the windows and measured it for four small tables. [6] To reduce noise without building walls, we suggested using bookcases as low dividers and attaching felt pads beneath the chairs. [7] The plan also includes a sign explaining that conversation should remain quiet. [8] These simple choices would give students another way to use a shared space without limiting the regular lunch area.

89. Which sentence should be removed because it does not help develop the proposal?

- A) Sentence 2
- B) Sentence 1
- C) Sentence 5
- D) Sentence 7

90. Which revision of sentence 4 is most effective?

- A) After observing the room, however, we noticed that carts passed that spot throughout lunch.
- B) After observing the room; however, we noticed, carts passed that spot throughout lunch.
- C) After observing the room, however, carts passing that spot throughout lunch.
- D) After observing the room however we noticed that carts, passed that spot throughout lunch.

91. Which phrase best replaces “simple choices” in sentence 8 to make the reference more precise?

- A) some ideas like that
- B) those kinds of things
- C) these changes
- D) everything mentioned before

92. Which sentence would provide the most effective conclusion after sentence 8?

- A) The kitchen doors are painted blue and have narrow windows.
- B) We recommend testing the corner for one month and gathering student feedback.
- C) Our council meets on the first Tuesday of each month.
- D) Some students bring sandwiches, while others purchase lunch.

Revision Passage: Learning from a Shadow

[1] For a science project, I tracked the shadow of a meter stick planted upright in our yard. [2] I measured the shadow at the same times on five clear days. [3] My first set of measurements were difficult to compare because I had rounded some lengths but not others. [4] I repeated the observations, recording every length to the nearest centimeter. [5] The shadow was longest early in the morning, shortened toward midday, and then lengthened again. [6] This pattern happened because the Sun appeared at different heights in the sky. [7] Although the project used simple tools, careful recording made the pattern easier to recognize.

93. Which revision corrects the error in sentence 3?

- A) My first set of measurements is difficult to compare because I had rounded some lengths but not others.
- B) My first set of measurements were difficult to compare because I has rounded some lengths but not others.
- C) My first set of measurements was difficult to compare because I had rounded some lengths but not others.
- D) My first set of measurements have been difficult to compare because I rounded some lengths but not others.

94. Which revision of sentence 5 uses parallel structure?

- A) The shadow was longest early in the morning, shortening toward midday, and then it lengthened again.
- B) The shadow being longest early, it shortened toward midday and lengthening again.
- C) The shadow was longest early in the morning, was shorter toward midday, and lengthening again.
- D) The shadow lengthened early in the morning, shortened toward midday, and lengthened again afterward.

95. Which revision of sentence 6 is most precise?

- A) The pattern was a pattern caused by changes that changed.
- B) The thing happened because the Sun did different stuff in the sky.
- C) This was due to it being somewhere else as time went by.
- D) The pattern occurred as the Sun's apparent height changed during the day.

96. Which sentence is written correctly?

- A) The award was presented to Maya and her.
- B) Neither of the lamps have a working bulb.
- C) Between you and I, the rehearsal needs more time.
- D) Each hikers carried their own water bottle.

97. Which sentence expresses the comparison most clearly and correctly?

- A) The new route, unlike the old route is being short.
- B) The new route is shorter compared to the old route was.
- C) Compared with the old route, the new one is more shorter.
- D) The new route is shorter than the old route.

98. Which numbered line contains an error in standard written US English?

Line 1: The collection of postcards belongs to my aunt.

Line 2: She stores them in a box with a tight lid.

Line 3: Several cards shows buildings that no longer exist.

Line 4: We examine one card whenever she visits.

- A) Line 1
- B) Line 2
- C) Line 4
- D) Line 3

99. Which numbered line contains an error in standard written US English?

Line 1: Before the doors opened, the musicians tuned their instruments.

Line 2: The conductor reviewed the first page of the score.

Line 3: Everyone waited quietly for the signal to begin.

Line 4: The audience take their seats as the lights dimmed.

- A) Line 1
- B) Line 2
- C) Line 4
- D) Line 3

100. Which numbered line contains an error in standard written US English?

Line 1: A row of young trees borders the path.

Line 2: Their branches provides shade during the summer.

Line 3: Visitors often rest on the benches nearby.

Line 4: The park crew waters the trees twice a week.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

Section 3: Mathematics

50 questions | Recommended time: 45 minutes

101. A number is divisible by both 6 and 15. Which is the least possible positive number?

- A) 30
- B) 60
- C) 21
- D) 90

102. Which value is greatest?

- A) 0.6
- B) 0.56
- C) 58%
- D) $\frac{7}{12}$

103. A shop mixes $2\frac{1}{4}$ kilograms of one type of grain with $1\frac{2}{3}$ kilograms of another. It packages the mixture equally into 6 bags. How much grain is in each bag?

- A) $\frac{7}{12}$ kilogram
- B) $\frac{23}{36}$ kilogram
- C) $\frac{13}{24}$ kilogram
- D) $\frac{47}{72}$ kilogram

104. A machine changes each input x according to the rule $4x + 3$. If the output is 35, what was the input?

- A) 8
- B) 9
- C) 7
- D) 11

105. The points (2, 11), (4, 17), and (6, 23) follow the same linear rule. Which equation gives y in terms of x ?

- A) $y = 2x + 7$
- B) $y = 4x + 3$
- C) $y = 3x + 5$
- D) $y = 6x - 1$

106. Which integer satisfies both inequalities $n > -4$ and $n = -1$?

- A) -5
- B) -2
- C) -4
- D) 0

107. Five delivery times, in minutes, are 18, 21, 21, 24, and 36. Which statement is true?

- A) The median is 21, and the mean is greater than the median.
- B) The mean and median are both 21.
- C) The range is 15, and the mode is 24.
- D) Removing 36 would increase the mean.

108. A spinner has 8 equal sections: 3 marked R, 2 marked S, and 3 marked T. It is spun twice. What is the probability of landing on S exactly once?

- A) $\frac{1}{8}$
- B) $\frac{1}{2}$
- C) $\frac{5}{8}$
- D) $\frac{3}{8}$

109. A regular polygon has an exterior angle of 45° at each vertex. How many sides does the polygon have?

- A) 6
- B) 8
- C) 9
- D) 7

110. A right triangle has legs 9 units and 12 units long. What is the length of its hypotenuse?

- A) 13 units
- B) 18 units
- C) 15 units
- D) 21 units

111. A rectangular storage box measures 40 centimeters by 25 centimeters by 18 centimeters. What is its volume in liters? Use 1,000 cubic centimeters = 1 liter.

- A) 1.8 liters
- B) 8 liters
- C) 180 liters
- D) 18 liters

112. On a scale drawing, 2.5 centimeters represents 6 meters. A wall measures 7.5 centimeters on the drawing. What is the wall's actual length?

- A) 12 meters
- B) 15 meters
- C) 18 meters
- D) 9 meters

113. What is the value of $\frac{5}{6} - \frac{3}{8}$?

- A) $\frac{11}{24}$
- B) $\frac{1}{2}$
- C) $\frac{7}{24}$
- D) $\frac{19}{48}$

114. A jacket's price increases from \$48 to \$60. What is the percent increase?

- A) 12%
- B) 20%
- C) 80%
- D) 25%

115. When positive integer n is divided by 7, the remainder is 4. Which could be the value of n ?

- A) 18
- B) 39
- C) 32
- D) 24

116. Solve $3(2x - 5) = 27$.

- A) 5
- B) 7
- C) 6
- D) 9

117. A taxi charge is represented by $c = 2.5m + 4$, where m is the number of miles traveled. What does 4 represent?

- A) The number of miles included at no cost
- B) The cost of traveling four miles
- C) The fixed charge before mileage is added
- D) The increase in cost for each mile

118. The numbers of books read by six students are 2, 3, 3, 5, 6, and 8. If another student who read 15 books is added, which measure changes the most?

- A) Median
- B) Mode
- C) Minimum
- D) Range

119. A bag contains 5 red counters, 4 blue counters, and 1 yellow counter. One counter is chosen at random. What is the probability that it is not blue?

- A) $\frac{1}{10}$
- B) $\frac{2}{5}$
- C) $\frac{3}{5}$
- D) $\frac{4}{5}$

120. A survey reports these preferences: walking 18, cycling 12, bus 9, other 6. What percent of the 45 respondents selected cycling?

- A) 20%
- B) $26\frac{2}{3}\%$
- C) 30%
- D) 36%

121. Two angles are supplementary. One angle measures 34° more than the other. What is the measure of the larger angle?

- A) 107°
- B) 73°
- C) 124°
- D) 146°

122. A circle has a diameter of 14 centimeters. Using $p = \frac{22}{7}$, what is its circumference?

- A) 22 centimeters
- B) 28 centimeters
- C) 44 centimeters
- D) 154 centimeters

123. A rectangular prism has length 8 units, width 5 units, and height 3 units. What is its surface area?

- A) 79 square units
- B) 120 square units
- C) 240 square units
- D) 158 square units

124. A 3.6-kilometer trail is marked every 300 meters. How many equal intervals are between the start and finish?

- A) 12
- B) 10
- C) 11
- D) 9

125. A pump fills 18 liters in 4 minutes at a constant rate. At that rate, how many liters does it fill in 10 minutes?

- A) 40 liters
- B) 45 liters
- C) 54 liters
- D) 72 liters

126. What is the prime factorization of 180?

- A) $2^2 \times 3 \times 15$
- B) $2 \times 3^2 \times 10$
- C) $2^3 \times 3 \times 5$
- D) $2^2 \times 3^2 \times 5$

127. A recipe uses flour and oats in the ratio 5:2. If 21 cups of the mixture are prepared in the same ratio, how many cups are oats?

- A) 3
- B) 6
- C) 5
- D) 15

128. Which expression has a value closest to 50?

- A) $196 \div 3.8$
- B) 7.8×5.9
- C) $24.6 + 21.7$
- D) $102.4 - 47.9$

129. The sequence begins 4, 9, 19, 39. Which rule generates each term after the first?

- A) Add consecutive odd numbers.
- B) Multiply by 3, then subtract 3.
- C) Multiply by 2, then add 1.
- D) Add 5, then add 10 repeatedly.

130. If $5x - 2y = 18$ and $x = 6$, what is the value of y ?

- A) 6
- B) 5
- C) 4
- D) 8

131. A theater has r rows with 18 seats in each row and 12 additional seats at the back. Which expression gives the total number of seats?

- A) $18(r + 12)$
- B) $12r + 18$
- C) $18r - 12$
- D) $18r + 12$

132. Four students read an average of 12 pages each, and six other students read an average of 17 pages each. What is the average number of pages read by all ten students?

- A) 15
- B) 14
- C) 14.5
- D) 16

133. A fair coin is tossed three times. What is the probability of getting at least one tail?

- A) $\frac{7}{8}$
- B) $\frac{1}{8}$
- C) $\frac{3}{4}$
- D) $\frac{1}{2}$

134. The area of a parallelogram is 96 square centimeters. Its base is 12 centimeters. What is its perpendicular height?

- A) 6 centimeters
- B) 12 centimeters
- C) 10 centimeters
- D) 8 centimeters

135. A square has the same perimeter as a rectangle measuring 10 units by 6 units. What is the area of the square?

- A) 32 square units
- B) 48 square units
- C) 80 square units
- D) 64 square units

136. A container holds 2.75 gallons. How many quarts does it hold? Use 1 gallon = 4 quarts.

- A) 6.75 quarts
- B) 8 quarts
- C) 11 quarts
- D) 9 quarts

137. A board is 7 feet 6 inches long. A 28-inch piece is cut from it. How many inches remain? Use 1 foot = 12 inches.

- A) 58 inches
- B) 50 inches
- C) 62 inches
- D) 72 inches

138. A recipe uses $2\frac{1}{4}$ cups of oats for 18 bars. How many cups are used per bar?

- A) $\frac{1}{16}$ cup
- B) $\frac{1}{2}$ cup
- C) $\frac{1}{4}$ cup
- D) $\frac{1}{8}$ cup

139. A data set is 6, 8, 8, 11, and x . If its mean is 9, what is x ?

- A) 12
- B) 10
- C) 11
- D) 9

140. A regular polygon has an exterior angle of 45° . How many sides does it have?

- A) 6
- B) 8
- C) 7
- D) 9

141. For the function $y = 3x - 4$, which ordered pair is on its graph?

- A) (1, 1)
- B) (3, 4)
- C) (2, 2)
- D) (4, 7)

142. A bus leaves at 8:47 a.m. and arrives at 11:23 a.m. How long is the trip?

- A) 2 hours 24 minutes
- B) 2 hours 36 minutes
- C) 3 hours 36 minutes
- D) 3 hours 24 minutes

143. Which value is greatest?

- A) 62%
- B) 0.61
- C) $\frac{5}{8}$
- D) 0.603

144. A bag holds 3 red, 2 blue, and 5 green tiles. Two tiles are drawn without replacement. What is the probability both are blue?

- A) $\frac{1}{10}$
- B) $\frac{1}{25}$
- C) $\frac{2}{45}$
- D) $\frac{1}{45}$

145. A right triangle has legs of 9 units and 12 units. What is the length of its hypotenuse?

- A) 13 units
- B) 14 units
- C) 21 units
- D) 15 units

146. Solve $5(x - 2) = 3x + 8$.

- A) 7
- B) 9
- C) 8
- D) 10

147. A map scale states that 1 inch represents 18 miles. Two towns are $2\frac{1}{2}$ inches apart on the map. What is their actual distance?

- A) 40 miles
- B) 36 miles
- C) 45 miles
- D) 54 miles

148. A price of \$72 is reduced by 25% and then increased by \$6. What is the final price?

- A) \$54
- B) \$66
- C) \$60
- D) \$72

149. Team A scored 12, 16, 18, and 22 points. Team B scored 14, 15, 19, and 20 points. Which statement is true?

- A) Team A has a greater mean and a smaller range.
- B) The teams have equal means, and Team A has the greater range.
- C) Team B has a greater mean and a greater range.
- D) The teams have equal means and equal ranges.

150. A rectangle has perimeter 34 units and length 11 units. What is its area?

- A) 55 square units
- B) 66 square units
- C) 60 square units
- D) 72 square units

Section 4: Ability

50 questions | Recommended time: 35 minutes

151. Complete the 2×2 matrix.

Top-left: ○ with + inside

Top-right: ● with + to its right

Bottom-left: □ with × inside

Bottom-right: ?

A) ■ with × to its right

B) □ with + to its right

C) ■ with + inside

D) ● with × below

152. Complete the 2×2 matrix.

Top-left: ↑ above ○

Top-right: → to the right of ○

Bottom-left: ← to the left of □

Bottom-right: ?

A) ↓ below □

B) → to the right of □

C) ↑ above □

D) ↓ below ○

153. Complete the 3×3 matrix.

Row 1: ○ | □ | ○□

Row 2: △ | ◇ | △◇

Row 3: ☆ | + | ?

A) ☆+

B) ☆☆

C) ☆■

D) +☆

154. Complete the 2×2 matrix.

Top-left: Top: ○; Bottom: ■

Top-right: Top: ■; Bottom: ○

Bottom-left: Top: △; Bottom: ◆

Bottom-right: ?

A) Top: △; Bottom: ◆

B) Top: ▲; Bottom: ◇

C) Top: ◇; Bottom: ▲

D) Top: ◆; Bottom: △

155. Complete the 3×3 matrix.

Row 1: + | × | +×

Row 2: × | - | ×-

Row 3, cell 1: -

Row 3, cell 2: |

Row 3, cell 3: ?

- A) -|
- B) --
- C) |-
- D) +|

156. A square sheet is folded in half from bottom to top. Two holes are punched: one directly on the folded horizontal edge near the left side and one away from that edge near the right side. When the sheet is completely unfolded, what appears?

- A) Two holes, both on the horizontal centerline
- B) Four holes forming a rectangle
- C) Three holes: one on the horizontal centerline at left and a vertical mirrored pair at right
- D) Three holes in one horizontal row

157. A square sheet is folded along the diagonal from the upper-left corner to the lower-right corner. The resulting triangle is then folded so the upper-right corner meets the lower-left corner. One hole is punched away from all folded edges. When the sheet is completely unfolded, what appears?

- A) Two holes mirrored across one diagonal only
- B) Three holes forming a triangle
- C) Four holes related by reflections across both diagonals
- D) Eight holes arranged around the center

158. A rectangular sheet is folded in half from right to left. A small V-shaped cut is made on the bottom edge of the folded sheet, away from the vertical fold. When the sheet is unfolded, what appears?

- A) One diamond-shaped opening at the center
- B) Two matching V-shaped notches on the bottom edge, mirrored left and right
- C) Two matching V-shaped notches on the left and right edges
- D) Four V-shaped notches along the bottom edge

159. A square sheet is folded from left to right and then from top to bottom. One hole is punched directly on the folded vertical edge but away from the folded horizontal edge. When the sheet is completely unfolded, what appears?

- A) One hole at the center
- B) Four holes, one in each quadrant
- C) Two holes on the horizontal centerline, reflected across the vertical centerline
- D) Two holes on the vertical centerline, reflected across the horizontal centerline

160. The first three arrangements follow the same rule.

○ + ●

□ + ■

△ + ▲

Which choice follows the same rule?

- A) ◇ + ■
- B) ◆ + ◇
- C) ☆ + ◆
- D) ◇ + ◆

161. The first three arrangements follow the same rule.

↑ → ↓
→ ↓ ←
↓ ← ↑

Which choice follows the same rule?

- A) ← ↓ →
- B) ← ↑ →
- C) ↑ ← ↓
- D) → ↑ ←

162. The first three figures follow the same rule.

○ inside □, with + outside
△ inside ◇, with × outside
☆ inside ○, with | outside

Which choice follows the same rule?

- A) ● inside □, with + outside
- B) □ inside △, with - outside
- C) □ inside △, with × inside
- D) □ inside □, with - outside

163. Complete the 3 × 3 matrix. In each row, the third cell contains the symbols that appear in only one of the first two cells.

Row 1: ○□ | □△ | ○△

Row 2: ☆+ | +× | ☆×

Row 3, cell 1: ◇|

Row 3, cell 2: |■

Row 3, cell 3: ?

- A) ◇|■
- B) |■
- C) ◇|
- D) ◇■

164. Complete the 2 × 2 matrix.

Top-left: ○ with a vertical line inside

Top-right: ○ with a diagonal line / inside

Bottom-left: ● with a horizontal line inside

Bottom-right: ?

- A) ○ with a diagonal line / inside
- B) ● with a horizontal line inside
- C) ● with a diagonal line \ inside
- D) ● with a vertical line inside

165. Complete the 3 × 3 matrix.

Row 1: ○ | ○○ | ○○○

Row 2: □□ | □□□ | □□□□

Row 3: △△△ | △△△△ | ?

- A) △△△
- B) △△△△△
- C) △△△△△△
- D) △△△△△△△

166. Complete the 2×2 matrix.

Top-left: + above ○

Top-right: ○ above +

Bottom-left: × to the left of □

Bottom-right: ?

- A) □ above ×
- B) × to the right of □
- C) + to the right of ○
- D) □ to the left of ×

167. A square sheet is folded in half from left to right. Two holes are punched: one directly on the vertical fold near the top and one away from the fold near the bottom. When the sheet is completely unfolded, what appears?

- A) Three holes: one on the vertical centerline near the top and a left-right mirrored pair near the bottom
- B) Two holes on the vertical centerline
- C) Four holes, one near each corner
- D) Three holes in one vertical row

168. A rectangular sheet is folded from bottom to top and then from left to right. A semicircular cut is made on the folded horizontal edge, away from the folded vertical edge. When the sheet is completely unfolded, what appears?

- A) Four complete circular openings, one in each quadrant
- B) Four separate semicircular notches on the outer edges
- C) Two complete circular openings mirrored across the vertical centerline
- D) One circular opening at the center

169. A square sheet is folded along the diagonal from the lower-left corner to the upper-right corner. One hole is punched directly on the diagonal fold, away from the center. When the sheet is unfolded, what appears?

- A) One hole lying on the diagonal
- B) Four holes around the center
- C) Two holes mirrored across the diagonal
- D) Two holes on the vertical centerline

170. A square sheet is folded from right to left, then from bottom to top, and finally along the diagonal of the folded square. One hole is punched away from every folded edge. When the sheet is completely unfolded, how many holes appear?

- A) 2
- B) 8
- C) 6
- D) 4

171. The first three arrangements follow the same rule.

○ ■ □

△ ◆ ◇

☆ ● ○

Which choice follows the same rule?

- A) ○ □ ●
- B) ◆ △ ◇
- C) ◇ ▲ □
- D) □ ■ □

172. The first three arrangements follow the same rule.

Top: ○; **Center:** +; **Bottom:** ○

Top: □; **Center:** ×; **Bottom:** □

Top: △; **Center:** |; **Bottom:** △

Which choice follows the same rule?

A) Top: ◇; Center: -; Bottom: ◇

B) Top: ◇; Center: ◇; Bottom: ◇

C) Top: ◇; Center: +; Bottom: ◆

D) Top: -; Center: ◇; Bottom: -

173. The first three arrangements follow the same rule.

↑ ○ →

→ □ ↓

↓ △ ←

Which choice follows the same rule?

A) ← ◇ ↓

B) ← ◇ ↑

C) → ○ ↑

D) ↑ ☆ ←

174. The first three figures follow the same rule.

Outer: ○; **Inner:** □; **symbol to the right:** +

Outer: △; **Inner:** ◇; **symbol to the right:** ×

Outer: ☆; **Inner:** ○; **symbol to the right:** |

Which choice follows the same rule?

A) Outer: ◆; Inner: △; symbol to the right: +

B) Outer: □; Inner: □; symbol to the right: -

C) Outer: □; Inner: △; symbol inside: ×

D) Outer: □; Inner: △; symbol to the right: -

175. Complete the 3 × 3 matrix. In each row, the third cell contains the first cell with its open shapes filled and its filled shapes opened.

Row 1: ○■ | + | ●□

Row 2: △◆ | × | ▲◇

Row 3: ☆● | - | ?

A) ★●

B) ☆○

C) ★○

D) ○★

176. Complete the 2 × 2 matrix.

Top-left: + to the left of ○

Top-right: + above ○

Bottom-left: × below □

Bottom-right: ?

A) × to the right of □

B) × above □

C) □ above ×

D) × to the left of □

177. Complete the 3×3 matrix.

Row 1: ○○○ | ●● | ○

Row 2: ■■■■ | □□ | ■■

Row 3: ▲▲▲▲▲ | △△△△ | ?

- A) ▲▲▲
- B) ▲▲▲▲▲
- C) △△
- D) △

178. Complete the 2×2 matrix.

Top-left: ↑ outside ○

Top-right: → inside ○

Bottom-left: ← outside ●

Bottom-right: ?

- A) ↓ outside ●
- B) → outside ○
- C) ↑ inside ●
- D) ↑ outside ●

179. Complete the 3×3 matrix. The third cell in each row shows only the symbol shared by the first two cells.

Row 1: ○□ | □△ | □

Row 2: ☆+ | +× | +

Row 3, cell 1: ◇■

Row 3, cell 2: ■|

Row 3, cell 3: ?

- A) ◇
- B) ■
- C) |
- D) ◇|

180. A square sheet is folded from top to bottom. Two holes are punched directly on the horizontal fold, one left of center and one right of center. When the sheet is unfolded, what appears?

- A) Two holes remaining on the horizontal centerline
- B) Four holes forming a rectangle
- C) Two vertical pairs of holes
- D) One hole at the center

181. A rectangular sheet is folded from left to right, then from bottom to top. One hole is punched directly on the folded vertical edge and directly on the folded horizontal edge, at their intersection. When the sheet is completely unfolded, what appears?

- A) Four holes around the center
- B) Two holes on the vertical centerline
- C) Two holes on the horizontal centerline
- D) One hole at the center of the original sheet

182. A square sheet is folded along the diagonal from upper-right to lower-left. A V-shaped cut is made on the folded diagonal edge, away from both corners. When the sheet is unfolded, what appears?

- A) Two separate V-shaped notches on outer edges
- B) Four triangular openings around the center
- C) One V-shaped notch at a corner
- D) One symmetric diamond-shaped opening centered on the diagonal

183. A square sheet is folded from bottom to top and then along the diagonal of the folded rectangle from its lower-left corner to its upper-right corner. One hole is punched away from every folded edge. When the sheet is completely unfolded, how many holes appear?

- A) 2
- B) 4
- C) 8
- D) 6

184. The first three arrangements follow the same rule.

○ | □□
 △ | ◇◇
 ☆ | ○○

Which choice follows the same rule?

- A) ◆ | △△
- B) □ | ◆◆
- C) ◇◇ | ◆
- D) ● | ■

185. The first three arrangements follow the same rule.

Top-left: ○; Top-right: □; Bottom-left: □; Bottom-right: ○
 Top-left: △; Top-right: ◇; Bottom-left: ◇; Bottom-right: △
 Top-left: ☆; Top-right: +; Bottom-left: +; Bottom-right: ☆
Which choice follows the same rule?

- A) Top-left: ●; Top-right: ■; Bottom-left: ●; Bottom-right: ■
- B) Top-left: □; Top-right: △; Bottom-left: □; Bottom-right: △
- C) Top-left: □; Top-right: □; Bottom-left: △; Bottom-right: △
- D) Top-left: ◇; Top-right: ○; Bottom-left: ○; Bottom-right: ◇

186. The first three arrangements follow the same rule.

↑ + ↓
 → × ←
 ↓ | ↑

Which choice follows the same rule?

- A) ← + ↓
- B) ↑ - →
- C) ← - →
- D) → | ↓

187. The first three figures follow the same rule.

Outer: □; Inner: ●; symbol below: +

Outer: ◇; Inner: ▲; symbol below: ×

Outer: ○; Inner: ★; symbol below: |

Which choice follows the same rule?

A) Outer: ■; Inner: ○; symbol below: +

B) Outer: ☆; Inner: ◆; symbol below: -

C) Outer: △; Inner: △; symbol below: ×

D) Outer: ○; Inner: ●; symbol inside: |

188. Complete the matrix. In each row, the right cell contains the left cell reflected left-to-right, with open shapes changed to filled shapes.

Row 1 left: Top-left ○; bottom-right □

Row 1 right: Top-right ●; bottom-left ■

Row 2 left: Top-left △; bottom-right ◇

Row 2 right: ?

A) Top-right ▲; bottom-left ◆

B) Top-left ▲; bottom-right ◆

C) Top-right △; bottom-left ◇

D) Top-left ◆; bottom-right ▲

189. Complete the matrix. Across each row, the second cell is the first cell rotated 90° clockwise, and the third is rotated another 90° clockwise.

Row 1: ↑ | → | ↓

Row 2: ← | ↑ | ?

A) ←

B) →

C) ↑

D) ↓

190. Complete the matrix. In each row, the third cell contains the symbols that appear in exactly one of the first two cells.

Row 1: {○, △} | {△, □} | {○, □}

Row 2: {★, +} | {+, ×} | ?

A) {★, +, ×}

B) {★, ×}

C) {+}

D) {★, +}

191. Complete the matrix. Down each column, the lower cell moves the symbol one corner clockwise and changes filled to open.

Top-left cell: Top-left ●

Bottom-left cell: Top-right ○

Top-right cell: Bottom-right ■

Bottom-right cell: ?

A) Bottom-left □

B) Top-right □

C) Bottom-right ■

D) Top-left □

192. A square paper is folded from right to left along the vertical centerline. It is then folded from bottom to top along the horizontal centerline. One hole is punched near the center of the folded square, away from every edge. What appears when the paper is fully unfolded?

- A) Two holes in the upper half
- B) Four holes along the vertical centerline
- C) Eight holes around the edges
- D) Four holes, one in each quadrant

193. A rectangular paper is folded from left to right along the vertical centerline. A semicircular cut is made on the folded edge, away from the top and bottom edges. What appears after unfolding?

- A) One semicircular cut on the left edge
- B) Two separate semicircular cuts on the outer edges
- C) One complete circular hole centered on the vertical centerline
- D) Four complete circular holes

194. A square paper is folded along the diagonal from upper-left to lower-right, bringing the upper-right corner onto the lower-left corner. One hole is punched near the folded triangle's upper-left corner, not on an edge. What appears after unfolding?

- A) One hole on the diagonal
- B) Four holes mirrored across both centerlines
- C) Two holes mirrored across that diagonal
- D) Eight holes arranged in a circle

195. A square is folded from top to bottom, then from left to right. A small triangular cut is made at the corner of the folded square where both folded edges meet. What appears after unfolding?

- A) One triangular hole at the center
- B) Two triangular holes on the top edge
- C) Four triangular cuts, one at each outer corner
- D) Four triangular cuts meeting symmetrically at the center

196. The following figures share one rule:

1. Outer ○; inner ●

2. Outer □; inner ■

3. Outer △; inner ▲

Which choice follows the same rule?

- A) Outer ◇; inner ◆
- B) Outer ◆; inner ◇
- C) Outer ◇; inner ◇
- D) Outer ★; inner ☆

197. The following arrangements share one rule:

1. Top ○; center □; bottom ○

2. Top △; center ★; bottom △

3. Top +; center ■; bottom +

Which choice follows the same rule?

- A) Top ◇; center ●; bottom ◇
- B) Top ◇; center ◇; bottom ●
- C) Top ●; center ●; bottom ○
- D) Top □; center △; bottom ○

198. The following sequences share one rule:

1. ○ ● ○ ●

2. □ ■ □ ■

3. △ ▲ △ ▲

Which choice follows the same rule?

A) ◇ ◇ ◆ ◆

B) ◇ ◆ ◇ ◆

C) ☆ ★ ☆ ★

D) ◆ ◇ ◆ ◇

199. The following arrangements share one rule:

1. Arrow sequence: ↑ → ↓

2. Arrow sequence: → ↓ ←

3. Arrow sequence: ↓ ← ↑

Which choice follows the same rule?

A) ← ↓ →

B) ← ↑ →

C) ↑ ← ↓

D) → ↑ ←

200. The following figures share one rule:

1. Two vertical lines inside □

2. Three vertical lines inside △

3. Four vertical lines inside ◇

The number of interior lines equals the number of the figure's corners minus 2. Which choice follows the rule?

A) Three lines inside □

B) Two lines inside △

C) One line inside △

D) Three lines inside ☆

Answer Key

Question numbers continue across all five sections.

Reading

1. B	2. D	3. B	4. C	5. D	6. B	7. D	8. A
9. C	10. D	11. B	12. C	13. A	14. D	15. A	16. D
17. C	18. B	19. A	20. D	21. A	22. C	23. B	24. B
25. D	26. A	27. C	28. D	29. D	30. A	31. A	32. B
33. C	34. A	35. A	36. C	37. A	38. B	39. D	40. C
41. B	42. D	43. A	44. B	45. B	46. C	47. A	48. C
49. C	50. A						

Written Expression

51. B	52. C	53. B	54. C	55. A	56. B	57. B	58. C
59. D	60. B	61. D	62. D	63. A	64. A	65. C	66. C
67. D	68. B	69. A	70. C	71. A	72. C	73. C	74. D
75. B	76. A	77. B	78. C	79. D	80. A	81. A	82. D
83. D	84. C	85. B	86. B	87. A	88. B	89. A	90. A
91. C	92. B	93. C	94. D	95. D	96. A	97. D	98. D
99. C	100. B						

Mathematics

101. A	102. A	103. D	104. A	105. C	106. B	107. A	108. D
109. B	110. C	111. D	112. C	113. A	114. D	115. B	116. B
117. C	118. D	119. C	120. B	121. A	122. C	123. D	124. A
125. B	126. D	127. B	128. A	129. C	130. A	131. D	132. A
133. A	134. D	135. D	136. C	137. C	138. D	139. A	140. B
141. C	142. B	143. C	144. D	145. D	146. B	147. C	148. C
149. B	150. B						

Ability

151. A	152. C	153. A	154. D	155. A	156. C	157. C	158. B
159. D	160. D	161. B	162. B	163. D	164. C	165. D	166. D
167. A	168. C	169. A	170. B	171. C	172. A	173. B	174. D
175. C	176. D	177. A	178. C	179. B	180. A	181. D	182. D
183. B	184. B	185. D	186. C	187. B	188. A	189. B	190. B
191. A	192. D	193. C	194. C	195. D	196. A	197. A	198. B

199. B	200. C						
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Detailed Explanations

Reading

1. TACHS-R5-001 - Answer B: The wind moves in different directions at different heights.
The low and high ribbons point differently, showing that wind direction changes with height.
2. TACHS-R5-002 - Answer D: He feels that changing the conditions might make his design seem unsuccessful.
Eli associates moving the launch line with admitting that his carefully built glider has failed.
3. TACHS-R5-003 - Answer B: Good problem solving includes examining conditions beyond the object that seems to be failing.
The successful solution comes from studying the surrounding wind rather than altering the glider.
4. TACHS-R5-004 - Answer C: placed quietly into.
Eli returns the unused clay to the box without drawing attention to the action.
5. TACHS-R5-005 - Answer D: Gaps slow the air without forcing all of it rapidly over the fence.
The passage explains that porous fences reduce wind speed while allowing air to pass through.
6. TACHS-R5-006 - Answer B: It explains how a tool begins a natural process, identifies design conditions, and describes limitations.
The passage moves from dune formation to fence placement, plant support, suitable locations, and limits.
7. TACHS-R5-007 - Answer D: To show one way managers protect plants that help stabilize dunes
Raised walkways keep visitors away from grasses whose roots and stems strengthen developing dunes.
8. TACHS-R5-008 - Answer A: A broad beach where steady winds carry dry sand inland
The passage says fences work best where wind moves dry sand inland and space permits dunes to form.
9. TACHS-R5-009 - Answer C: Effective interviewing balances preparation with attentive, respectful pauses.
The passage values prepared questions while explaining how pauses can invite deeper and more accurate responses.
10. TACHS-R5-010 - Answer D: The speaker is searching for words.
The passage directly identifies these relaxed signals as possible evidence of reflection.
11. TACHS-R5-011 - Answer B: a basic structure that guides the conversation.
Prepared questions organize the interview without controlling every moment, so they function as a guiding structure.
12. TACHS-R5-012 - Answer C: The interviewer may collect facts while missing deeper reflections.
Rushing to the next question can prevent unexpected memories and meanings from emerging.
13. TACHS-R5-013 - Answer A: practical and thoughtful.
The author offers usable guidance while carefully considering both reflection and discomfort during pauses.
14. TACHS-R5-014 - Answer D: The suitcase lid sends the hat onto Lila's head.
The lid springs open and sends Andre's hat directly onto Lila's head.
15. TACHS-R5-015 - Answer A: Too much is explained verbally instead of shown efficiently.
The actors later replace three explanatory lines with a brief visual sequence that communicates the confusion faster.
16. TACHS-R5-016 - Answer D: Shared creativity can strengthen a leader's original idea.
Tessa keeps her story while allowing the actors' contributions to make the scene clearer and more efficient.
17. TACHS-R5-017 - Answer C: included and communicated.
The finished scene contains and expresses ideas contributed by other group members.
18. TACHS-R5-018 - Answer B: A crack forming or extending as stress is released
Rapid cooling builds stress until cracking releases energy as vibrations through the ice and water.
19. TACHS-R5-019 - Answer A: It describes a phenomenon, explains its cause, corrects a possible misunderstanding, and notes a research use.
The paragraphs move from the sounds to their cause, safety limits, and scientific study.
20. TACHS-R5-020 - Answer D: To emphasize that sound is not a dependable measure of safety
The contrast shows that loudness and silence do not reliably indicate ice strength.
21. TACHS-R5-021 - Answer A: Temperature-driven cracks can make ice sound unusual, but the sounds do not establish whether the ice is safe.
This choice includes both the physical explanation and the passage's safety qualification.
22. TACHS-R5-022 - Answer C: To show the contents without breaking the sealed container
Exterior impressions allowed people to identify the enclosed tokens after the clay hardened.

23. TACHS-R5-023 - Answer B: continuing to exist from an earlier time.
The word describes ancient objects that remain available for scholars to study today.
24. TACHS-R5-024 - Answer B: Information became less dependent on storing the represented objects themselves.
Marks on tablets preserved information without requiring tokens to be sealed inside containers.
25. TACHS-R5-025 - Answer D: acknowledge that conclusions are based on incomplete evidence.
The author qualifies scholarly interpretations because future discoveries may change current understanding.
26. TACHS-R5-026 - Answer A: A bright strip of removable tape
Nora marks June's faded acceleration point with a visible strip of removable tape.
27. TACHS-R5-027 - Answer C: She has carefully studied the team's handoffs during practice.
Her observations have already taught her the marks, calls, and timing used by the runners.
28. TACHS-R5-028 - Answer D: Careful preparation can make a contribution valuable before it becomes visible.
Nora's observation work improves the team and prepares her to run when needed.
29. TACHS-R5-029 - Answer D: began participating directly in.
"Entered" refers to Nora physically joining the relay lineup, not merely recording information.
30. TACHS-R5-030 - Answer A: The puzzle is clear but not very challenging.
The tester reaches the goal immediately and says the activity does not feel like much of a puzzle.
31. TACHS-R5-031 - Answer A: She now understands that mistakes can reveal useful possibilities.
The discarded routes inspired optional paths, changing Priya's view of failed attempts.
32. TACHS-R5-032 - Answer B: the revised puzzle supports more than one satisfying way to play.
Visitors can pursue speed or collect every token, demonstrating the revised design's flexibility.
33. TACHS-R5-033 - Answer C: It creates a problem that leads the designers to reconsider their discarded attempts.
The tester exposes the lack of challenge, prompting Mateo and Priya to use their failed routes.
34. TACHS-R5-034 - Answer A: Gravity
Once droplets combine and become heavy enough, gravity pulls them into the channel below.
35. TACHS-R5-035 - Answer A: how effectively the net collects water.
The paragraph discusses how mesh design and placement affect the amount of water captured.
36. TACHS-R5-036 - Answer C: Fog nets collect airborne droplets through carefully positioned mesh, but their usefulness depends on local conditions.
This choice includes both the collection process and the passage's important limits.
37. TACHS-R5-037 - Answer A: An exposed ridge facing regular fog-bearing winds
The passage says orientation toward frequent moist winds strongly improves collection.
38. TACHS-R5-038 - Answer B: To clarify a limitation and prevent an exaggerated understanding of the technology
The statement emphasizes that fog nets capture existing droplets rather than producing a new water supply.
39. TACHS-R5-039 - Answer D: A short hiss from a spray bottle
The spray-bottle hiss joins the clicking spokes to imitate a train gathering speed.
40. TACHS-R5-040 - Answer C: She wants to leave room for another possible use.
Pencil suggests the label can change, matching Lena's realization that objects may have multiple uses.
41. TACHS-R5-041 - Answer B: the effectiveness of Lena's inventive backstage work.
The unseen wheel successfully creates a convincing effect, demonstrating Lena's creative contribution.
42. TACHS-R5-042 - Answer D: lacking realism or energy.
The recorded noise sounds unconvincing and lifeless, so "flat" means lacking realism or energy.
43. TACHS-R5-043 - Answer A: Repeated freezing and thawing
The passage directly names repeated freezing and thawing as one process that can weaken a hard coat.
44. TACHS-R5-044 - Answer B: Seed dormancy delays growth until particular signals improve the chance of survival.
The passage explains several dormancy signals and shows how delayed germination can improve survival.
45. TACHS-R5-045 - Answer B: suitable for success.
The paragraph contrasts destructive frost with conditions suitable for successful growth.
46. TACHS-R5-046 - Answer C: To illustrate the risk of an entire group sprouting at once
The late-frost example demonstrates the danger of simultaneous germination and the advantage of spreading risk.
47. TACHS-R5-047 - Answer A: A curb cut collects rainwater.
The parent specifically points out that rainwater gathers at a curb cut.

48. TACHS-R5-048 - Answer C: It is useful but does not capture every relevant experience.
Reed retains traffic counts while supplementing them with observations, showing she values both sources.
49. TACHS-R5-049 - Answer C: Combining numerical records with users' experiences can lead to better design.
Reed's successful changes result from combining conventional data with the experiences of varied users.
50. TACHS-R5-050 - Answer A: A problem is identified, additional evidence is gathered, and solutions are applied.
The passage moves from an incomplete understanding to new evidence, proposed changes, and observed results.

Written Expression

51. TACHS-WE5-001 - Answer B: Sentence 3

Basketball practice is unrelated to planning an audio guide for the history display.

52. TACHS-WE5-002 - Answer C: Students will record their scripts in the media room, which has sound-absorbing panels. The relative clause correctly joins the related ideas and removes the comma splice.

53. TACHS-WE5-003 - Answer B: Each recording should give visitors a focused explanation of one object in about a minute. This revision states the recording's purpose precisely and naturally without vague wording.

54. TACHS-WE5-004 - Answer C: For example, a speaker describing an old lunch token might explain what it cost and point out the faded school emblem.

The example directly develops the three recommended elements of an effective object description.

55. TACHS-WE5-005 - Answer A: Pulling the thread as hard as possible, I tore one of my pages.

The introductory phrase correctly modifies "I," the person who pulled the thread.

56. TACHS-WE5-006 - Answer B: My first attempt failed.

The revision removes redundant wording while preserving the original meaning.

57. TACHS-WE5-007 - Answer B: After sentence 3

The added sentence explains a useful property of the waxed thread introduced in sentence 3.

58. TACHS-WE5-008 - Answer C: On my second attempt, I tightened each stitch only until the pages rested firmly together; as a result, the cover lay flat and the pages turned without catching.

This choice joins the complete ideas correctly and makes their cause-and-effect relationship clear.

59. TACHS-WE5-009 - Answer D: By the time the bus arrived, we had waited for nearly thirty minutes.

Past perfect correctly shows that the waiting occurred before the bus arrived.

60. TACHS-WE5-010 - Answer B: Of the two routes, the path beside the canal is shorter.

The comparative "shorter" is correct when comparing exactly two routes.

61. TACHS-WE5-011 - Answer D: Would you please let me know the hours for the fossil gallery during spring break?

The sentence uses courteous, specific language appropriate for a formal request.

62. TACHS-WE5-012 - Answer D: Line 3

The plural subject "Several" requires the plural verb "have," not "has."

63. TACHS-WE5-013 - Answer A: Line 4

The verb "laid" should be modified by the adverb "carefully," not the adjective "careful."

64. TACHS-WE5-014 - Answer A: The sand filter produced clearer water, but it required nearly twice as long to drain.

The coordinating conjunction correctly joins the contrasting independent clauses and removes the comma splice.

65. TACHS-WE5-015 - Answer C: Sentence 6

Stating that the cups were on a table does not develop the comparison or explain the results.

66. TACHS-WE5-016 - Answer C: Neither filter purified the water for drinking; the test measured only filtration speed and visible clarity.

The revision states the experiment's safety limitation and measured variables precisely.

67. TACHS-WE5-017 - Answer D: This trade-off explains why the clearer sample took longer to collect.

The sentence directly connects the proposed explanation with both observed results.

68. TACHS-WE5-018 - Answer B: After choosing the poems, the committee would map the locations of the stands.

The revision gives the introductory modifier a clear subject and uses standard active construction.

69. TACHS-WE5-019 - Answer A: Visitors could explore the stands in any order and pause at appealing titles.

This revision preserves both ideas in direct, concise language.

70. TACHS-WE5-020 - Answer C: The path would turn an ordinary crossing place into a brief, shared encounter with student writing.

The sentence concludes by emphasizing the proposal's purpose and effect on its audience.

71. TACHS-WE5-021 - Answer A: The display would remain for one week; then all stands would be removed so the courtyard could promptly return to regular use.

The revision combines the ideas with correct punctuation and a clear time-and-purpose relationship.

72. TACHS-WE5-022 - Answer C: Every one of the lanterns needs a fresh battery.

The singular subject "Every one" requires the singular verb "needs."

73. TACHS-WE5-023 - Answer C: The guide gave Omar and me a copy of the schedule.
"Omar and me" correctly functions as the indirect object of "gave."
74. TACHS-WE5-024 - Answer D: Line 4
The comparative adverb phrase should be "more neatly," not "more neat."
75. TACHS-WE5-025 - Answer B: Line 1
With "neither...nor," the verb agrees with the nearer singular subject "rain," so "was expected" is required.
76. TACHS-WE5-026 - Answer A: Line 3
The plural subject "two assistants" requires "carry," not "carries."
77. TACHS-WE5-027 - Answer B: At the fountain, splashing water and nearby conversations produced steady background noise.
The revision replaces vague language with specific, observable sources of sound.
78. TACHS-WE5-028 - Answer C: Sentence 7
The number of sinks in the art room is unrelated to measuring and applying sound data.
79. TACHS-WE5-029 - Answer D: At six locations, we recorded the loudest sound every five minutes during lunch and classified its source as people, traffic, equipment, or animals.
The sentence combines both actions with correct coordination and clear wording.
80. TACHS-WE5-030 - Answer A: The map can guide decisions about where to schedule quiet study and louder group events.
This revision states a specific decision the sound map can support.
81. TACHS-WE5-031 - Answer A: We used a ruler to guide each straight edge as we cut the openings with a craft knife.
The revision gives the introductory action a logical performer and clearly connects the tools to their uses.
82. TACHS-WE5-032 - Answer D: From the audience's viewpoint, that arrangement made the two entrances seem to lead to one room.
The sentence explains why the original door placement created a design problem.
83. TACHS-WE5-033 - Answer D: The revision improved the model.
The choice removes repetition and expresses the original meaning directly.
84. TACHS-WE5-034 - Answer C: The committee has chosen three photographs for the exhibit.
"Has chosen" uses the correct past participle with the helping verb.
85. TACHS-WE5-035 - Answer B: The oak tree lost several branches during the storm, but it remained standing.
The pronoun "it" clearly and logically refers to the singular oak tree.
86. TACHS-WE5-036 - Answer B: After the final bus arrived, the volunteers locked the gate.
This choice contains a dependent introductory clause followed by a complete independent clause.
87. TACHS-WE5-037 - Answer A: Line 4
The singular subject "guide" requires the present-tense verb "explains," not "explain."
88. TACHS-WE5-038 - Answer B: Line 1
The completed action before another past event requires "had loaded," not "has loaded."
89. TACHS-WE5-039 - Answer A: Sentence 2
Sentence 2 gives a general size fact that does not explain the need, location, or design of the quiet corner.
90. TACHS-WE5-040 - Answer A: After observing the room, however, we noticed that carts passed that spot throughout lunch.
Adding "that" creates a complete, smoothly punctuated sentence with a clear subordinate opening.
91. TACHS-WE5-041 - Answer C: these changes
"These changes" concisely and clearly refers to the proposed location, dividers, chair pads, and sign.
92. TACHS-WE5-042 - Answer B: We recommend testing the corner for one month and gathering student feedback.
A trial period and feedback provide a logical next step for evaluating the proposal.
93. TACHS-WE5-043 - Answer C: My first set of measurements was difficult to compare because I had rounded some lengths but not others.
The singular subject "set" requires the singular past-tense verb "was."
94. TACHS-WE5-044 - Answer D: The shadow lengthened early in the morning, shortened toward midday, and lengthened again afterward.
Choice D uses three coordinated past-tense verbs in a consistent parallel series.
95. TACHS-WE5-045 - Answer D: The pattern occurred as the Sun's apparent height changed during the day.
The revision precisely identifies the relevant change while avoiding vague and repetitive language.

96. TACHS-WE5-046 - Answer A: The award was presented to Maya and her.
“Her” is the correct object pronoun after “to,” and the sentence has standard agreement.
97. TACHS-WE5-047 - Answer D: The new route is shorter than the old route.
Choice D makes a direct, grammatical comparison without redundancy or a misplaced modifier.
98. TACHS-WE5-048 - Answer D: Line 3
Line 3 has a subject-verb agreement error; the plural subject “cards” requires “show.”
99. TACHS-WE5-049 - Answer C: Line 4
Line 4 shifts incorrectly to present tense; “took” should match the passage’s past tense.
100. TACHS-WE5-050 - Answer B: Line 2
Line 2 has a subject-verb agreement error; the plural subject “branches” requires “provide.”

Mathematics

101. TACHS-M5-001 - Answer A: 30

The least common multiple of 6 and 15 is 30.

102. TACHS-M5-002 - Answer A: 0.6

The values are about 0.583, 0.56, 0.58, and 0.6, so 0.6 is greatest.

103. TACHS-M5-003 - Answer D: 47/72 kilogram

$2\frac{1}{4} + 1\frac{2}{3} = 47/12$, and $47/12 \div 6 = 47/72$.

104. TACHS-M5-004 - Answer A: 8

Solve $4x + 3 = 35$: subtract 3 and divide by 4 to obtain $x = 8$.

105. TACHS-M5-005 - Answer C: $y = 3x + 5$

Each 2-unit increase in x raises y by 6, giving slope 3; substituting (2, 11) gives intercept 5.

106. TACHS-M5-006 - Answer B: -2

Integers greater than -4 and no greater than -1 include -3, -2, and -1.

107. TACHS-M5-007 - Answer A: The median is 21, and the mean is greater than the median.

The median is 21, while the mean is $120 \div 5 = 24$.

108. TACHS-M5-008 - Answer D: $3/8$

Exactly one S has probability $2 \times (2/8 \times 6/8) = 24/64 = 3/8$.

109. TACHS-M5-009 - Answer B: 8

Exterior angles total 360° , so the polygon has $360 \div 45 = 8$ sides.

110. TACHS-M5-010 - Answer C: 15 units

By the Pythagorean relationship, $\sqrt{9^2 + 12^2} = \sqrt{225} = 15$.

111. TACHS-M5-011 - Answer D: 18 liters

The volume is $40 \times 25 \times 18 = 18,000$ cubic centimeters, which equals 18 liters.

112. TACHS-M5-012 - Answer C: 18 meters

Since 7.5 centimeters is three times 2.5 centimeters, the actual length is $3 \times 6 = 18$ meters.

113. TACHS-M5-013 - Answer A: $11/24$

Using denominator 24 gives $20/24 - 9/24 = 11/24$.

114. TACHS-M5-014 - Answer D: 25%

The increase is \$12, and $12 \div 48 = 0.25$, or 25%.

115. TACHS-M5-015 - Answer B: 39

Since $39 = 7 \times 5 + 4$, dividing by 7 leaves remainder 4.

116. TACHS-M5-016 - Answer B: 7

Divide by 3 to get $2x - 5 = 9$; then $2x = 14$ and $x = 7$.

117. TACHS-M5-017 - Answer C: The fixed charge before mileage is added

In $c = 2.5m + 4$, the constant 4 is charged even when m equals zero.

118. TACHS-M5-018 - Answer D: Range

The original range is $8 - 2 = 6$; adding 15 changes it to $15 - 2 = 13$, a change of 7.

119. TACHS-M5-019 - Answer C: $3/5$

Six of the ten counters are not blue, so the probability is $6/10 = 3/5$.

120. TACHS-M5-020 - Answer B: $26\frac{2}{3}\%$

Cycling represents $12/45 = 4/15 = 26\frac{2}{3}\%$.

121. TACHS-M5-021 - Answer A: 107°

If the smaller angle is x , then $x + (x + 34) = 180$, so the larger angle is 107° .

122. TACHS-M5-022 - Answer C: 44 centimeters

Circumference equals $pd = 22/7 \times 14 = 44$ centimeters.

123. TACHS-M5-023 - Answer D: 158 square units

Surface area is $2(8 \times 5 + 8 \times 3 + 5 \times 3) = 2(79) = 158$.

124. TACHS-M5-024 - Answer A: 12

Convert 3.6 kilometers to 3,600 meters, then calculate $3,600 \div 300 = 12$ intervals.

125. TACHS-M5-025 - Answer B: 45 liters
The rate is $18 \div 4 = 4.5$ liters per minute, and $4.5 \times 10 = 45$.
126. TACHS-M5-026 - Answer D: $2^2 \times 3^2 \times 5$
 $180 = 18 \times 10 = (2 \times 3^2)(2 \times 5) = 2^2 \times 3^2 \times 5$.
127. TACHS-M5-027 - Answer B: 6
Oats are 2 of 7 ratio parts, so $21 \times 2/7 = 6$ cups.
128. TACHS-M5-028 - Answer A: $196 \div 3.8$
The estimates are about 48, 52, 46, and 55; approximately 52 is closest to 50.
129. TACHS-M5-029 - Answer C: Multiply by 2, then add 1.
Each term doubles and increases by 1: 4 to 9, 9 to 19, and 19 to 39.
130. TACHS-M5-030 - Answer A: 6
Substitute $x = 6$: $30 - 2y = 18$, so $2y = 12$ and $y = 6$.
131. TACHS-M5-031 - Answer D: $18r + 12$
The rows contain $18r$ seats altogether, and the 12 back seats are then added.
132. TACHS-M5-032 - Answer A: 15
The groups read $4 \times 12 + 6 \times 17 = 150$ pages altogether; $150 \div 10 = 15$.
133. TACHS-M5-033 - Answer A: $7/8$
Only TTT has no tail, with probability $1/8$, so the complement is $1 - 1/8 = 7/8$.
134. TACHS-M5-034 - Answer D: 8 centimeters
Since area = base $\times$ height, the height is $96 \div 12 = 8$ centimeters.
135. TACHS-M5-035 - Answer D: 64 square units
The rectangle's perimeter is 32, so each square side is 8 and its area is $8^2 = 64$.
136. TACHS-M5-036 - Answer C: 11 quarts
Multiply 2.75 gallons by 4 quarts per gallon to obtain 11 quarts.
137. TACHS-M5-037 - Answer C: 62 inches
Convert 7 feet 6 inches to 90 inches, then subtract 28 to obtain 62 inches.
138. TACHS-M5-038 - Answer D: $1/8$ cup
 $2 \frac{1}{4} = 9/4$, and $9/4 \div 18 = 9/72 = 1/8$.
139. TACHS-M5-039 - Answer A: 12
A mean of 9 requires a total of 45; $45 - (6 + 8 + 8 + 11) = 12$.
140. TACHS-M5-040 - Answer B: 8
The exterior angles total 360° , so the polygon has $360 \div 45 = 8$ sides.
141. TACHS-M5-041 - Answer C: (2, 2)
When $x = 2$, $y = 3(2) - 4 = 2$, so (2, 2) is on the graph.
142. TACHS-M5-042 - Answer B: 2 hours 36 minutes
From 8:47 to 9:00 is 13 minutes, then 2 hours 23 minutes more, totaling 2 hours 36 minutes.
143. TACHS-M5-043 - Answer C: $5/8$
 $5/8 = 0.625$, which is greater than 0.62, 0.61, and 0.603.
144. TACHS-M5-044 - Answer D: $1/45$
The probability is $2/10 \times 1/9 = 2/90 = 1/45$.
145. TACHS-M5-045 - Answer D: 15 units
By the Pythagorean relationship, $\sqrt{(9^2 + 12^2)} = \sqrt{225} = 15$.
146. TACHS-M5-046 - Answer B: 9
 $5x - 10 = 3x + 8$, so $2x = 18$ and $x = 9$.
147. TACHS-M5-047 - Answer C: 45 miles
Multiply 18 miles per inch by 2.5 inches to get 45 miles.
148. TACHS-M5-048 - Answer C: \$60
A 25% reduction is \$18, so $\$72 - \$18 + \$6 = \60 .
149. TACHS-M5-049 - Answer B: The teams have equal means, and Team A has the greater range.
Both totals are 68, so both means are 17; Team A's range is 10 and Team B's is 6.

150. TACHS-M5-050 - Answer B: 66 square units

$2(11 + w) = 34$ gives $w = 6$, so the area is $11 \times 6 = 66$ square units.

Ability

151. TACHS-A5-001 - Answer A: ■ with × to its right

Moving right fills the outer shape and moves the inner sign outside; moving down changes circle/+ to square/×.

152. TACHS-A5-002 - Answer C: ↑ above □

Across each row the arrow rotates 90° clockwise, while moving down rotates it 90° counterclockwise and changes ○ to □.

153. TACHS-A5-003 - Answer A: ☆+

In every row, the third cell joins the first cell followed by the second without changing either form.

154. TACHS-A5-004 - Answer D: Top: ◆; Bottom: △

Across each row the top and bottom figures exchange positions while retaining their original shapes and fill.

155. TACHS-A5-005 - Answer A: -|

Each row's third cell places the first symbol before the second symbol.

156. TACHS-A5-006 - Answer C: Three holes: one on the horizontal centerline at left and a vertical mirrored pair at right

The punch on the fold remains single, while the off-fold punch reflects across the horizontal centerline.

157. TACHS-A5-007 - Answer C: Four holes related by reflections across both diagonals

Two diagonal folds create four layers, so reversing both folds reflects the punch across each diagonal.

158. TACHS-A5-008 - Answer B: Two matching V-shaped notches on the bottom edge, mirrored left and right

The vertical half-fold reflects the bottom-edge cut to the corresponding position on the other half.

159. TACHS-A5-009 - Answer D: Two holes on the vertical centerline, reflected across the horizontal centerline

The punch lies on the vertical fold and remains there, while reversing the horizontal fold creates one reflected partner.

160. TACHS-A5-010 - Answer D: ◇ + ◆

Each valid pair shows the open form of one shape followed by the filled form of that same shape.

161. TACHS-A5-011 - Answer B: ← ↑ →

Each arrow in a valid arrangement is the preceding arrow rotated 90° clockwise.

162. TACHS-A5-012 - Answer B: □ inside △, with - outside

Each valid figure places one open shape inside a different open shape and a line symbol outside.

163. TACHS-A5-013 - Answer D: ◇■

The shared vertical line cancels, leaving the diamond from the first cell and the filled square from the second.

164. TACHS-A5-014 - Answer C: ● with a diagonal line \ inside

Moving right rotates the line 45° clockwise, while moving down fills the circle and rotates the line 90° clockwise.

165. TACHS-A5-015 - Answer D: △△△△△△△

In each row, the third count equals the sum of the first two counts: 3 + 4 = 7.

166. TACHS-A5-016 - Answer D: □ to the left of ×

Across each row the two symbols exchange positions; the lower row uses a horizontal arrangement instead of a vertical one.

167. TACHS-A5-017 - Answer A: Three holes: one on the vertical centerline near the top and a left-right mirrored pair near the bottom

The fold-line punch remains single, while the off-fold punch reflects across the vertical centerline.

168. TACHS-A5-018 - Answer C: Two complete circular openings mirrored across the vertical centerline

The cut on the horizontal fold joins into a circle, and reversing the vertical fold creates its mirrored partner.

169. TACHS-A5-019 - Answer A: One hole lying on the diagonal

A punch directly on the fold line maps onto itself when the single diagonal fold is reversed.

170. TACHS-A5-020 - Answer B: 8

Three folds create eight layers, so an off-edge punch appears in eight reflected positions.

171. TACHS-A5-021 - Answer C: ◇ ▲ □

Each valid arrangement has different open outer shapes with one filled shape in the center.

172. TACHS-A5-022 - Answer A: Top: ◇; Center: -; Bottom: ◇

Each valid arrangement has matching open shapes above and below a line symbol.

173. TACHS-A5-023 - Answer B: ← ◇ ↑

The right arrow is the left arrow rotated 90° clockwise, with an open shape between them.

174. TACHS-A5-024 - Answer D: Outer: □; Inner: △; symbol to the right: -

Each valid figure has two different open shapes nested together and a line symbol outside on the right.

175. TACHS-A5-025 - Answer C: ★○
The star changes from open to filled, and the circle changes from filled to open.
176. TACHS-A5-026 - Answer D: × to the left of □
Moving right shifts the first symbol 90° clockwise around the shape; moving down changes +/○ to ×/□ and reverses its side.
177. TACHS-A5-027 - Answer A: ▲ ▲ ▲
Each row decreases by one shape while alternating fill, and each new row begins with one additional filled or open shape.
178. TACHS-A5-028 - Answer C: ↑ inside ●
Moving right rotates the arrow 90° clockwise and moves it inside; moving down fills the circle and rotates the arrow 90° counterclockwise.
179. TACHS-A5-029 - Answer B: ■
The filled square is the only symbol appearing in both of the first two cells.
180. TACHS-A5-030 - Answer A: Two holes remaining on the horizontal centerline
Both punches lie on the fold and therefore map onto themselves when the sheet is opened.
181. TACHS-A5-031 - Answer D: One hole at the center of the original sheet
A punch at the intersection of both fold lines remains fixed under both reflections.
182. TACHS-A5-032 - Answer D: One symmetric diamond-shaped opening centered on the diagonal
The V-shaped cut reflects across the folded diagonal, joining with its mirror to form a diamond.
183. TACHS-A5-033 - Answer B: 4
Two folds create four layers, so an off-edge punch appears in four reflected positions.
184. TACHS-A5-034 - Answer B: □ | ◆◆
Each valid arrangement has one open shape followed by two identical filled shapes of a different type.
185. TACHS-A5-035 - Answer D: Top-left: ◇; Top-right: ○; Bottom-left: ○; Bottom-right: ◇
Each valid arrangement has matching symbols on both diagonals, producing 180° rotational symmetry.
186. TACHS-A5-036 - Answer C: ← - - →
The outer arrows point in opposite directions with one line symbol between them.
187. TACHS-A5-037 - Answer B: Outer: ☆; Inner: ◆; symbol below: -
Each valid figure places a filled shape inside a different open shape with a line symbol below.
188. TACHS-A5-038 - Answer A: Top-right ▲; bottom-left ◆
Reflecting left-to-right swaps the positions, and filling the shapes changes △ to ▲ and ◇ to ◆.
189. TACHS-A5-039 - Answer B: →
Starting with ←, two 90° clockwise rotations produce ↑ and then →.
190. TACHS-A5-040 - Answer B: {★, ×}
The shared + is removed, leaving the symbols present in only one set: ★ and ×.
191. TACHS-A5-041 - Answer A: Bottom-left □
Moving one corner clockwise from bottom-right gives bottom-left, and filled ■ changes to open □.
192. TACHS-A5-042 - Answer D: Four holes, one in each quadrant
Two perpendicular center folds reflect the punch into four corresponding positions, one in each quadrant.
193. TACHS-A5-043 - Answer C: One complete circular hole centered on the vertical centerline
The semicircle on the fold reflects across the centerline to form one complete circular hole.
194. TACHS-A5-044 - Answer C: Two holes mirrored across that diagonal
A single diagonal fold produces two holes at reflected locations on opposite sides of the fold line.
195. TACHS-A5-045 - Answer D: Four triangular cuts meeting symmetrically at the center
The corner formed by both fold lines becomes the paper's center, and both unfolds create four symmetric cuts there.
196. TACHS-A5-046 - Answer A: Outer ◇; inner ◆
Each figure has an open outer shape containing the filled version of the same shape.
197. TACHS-A5-047 - Answer A: Top ◇; center ●; bottom ◇
The top and bottom symbols match, while the center symbol is different.
198. TACHS-A5-048 - Answer B: ◇ ◆ ◇ ◆
Each sequence alternates the open and filled forms of the same shape, beginning with open.
199. TACHS-A5-049 - Answer B: ← ↑ →
Each arrow turns 90° clockwise from the preceding arrow; ←, ↑, → follows that rule.

200. TACHS-A5-050 - Answer C: One line inside $\triangle$
A triangle has three corners, so it must contain $3 - 2 = 1$ interior line.